

Autodesk® Scaleform®

Scaleform 4.3

/

(Scaleform 4.3 Mobile Game Kit)

Scaleform

Nate Mitchell

1.01

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Autodesk® Scaleform® 4.3

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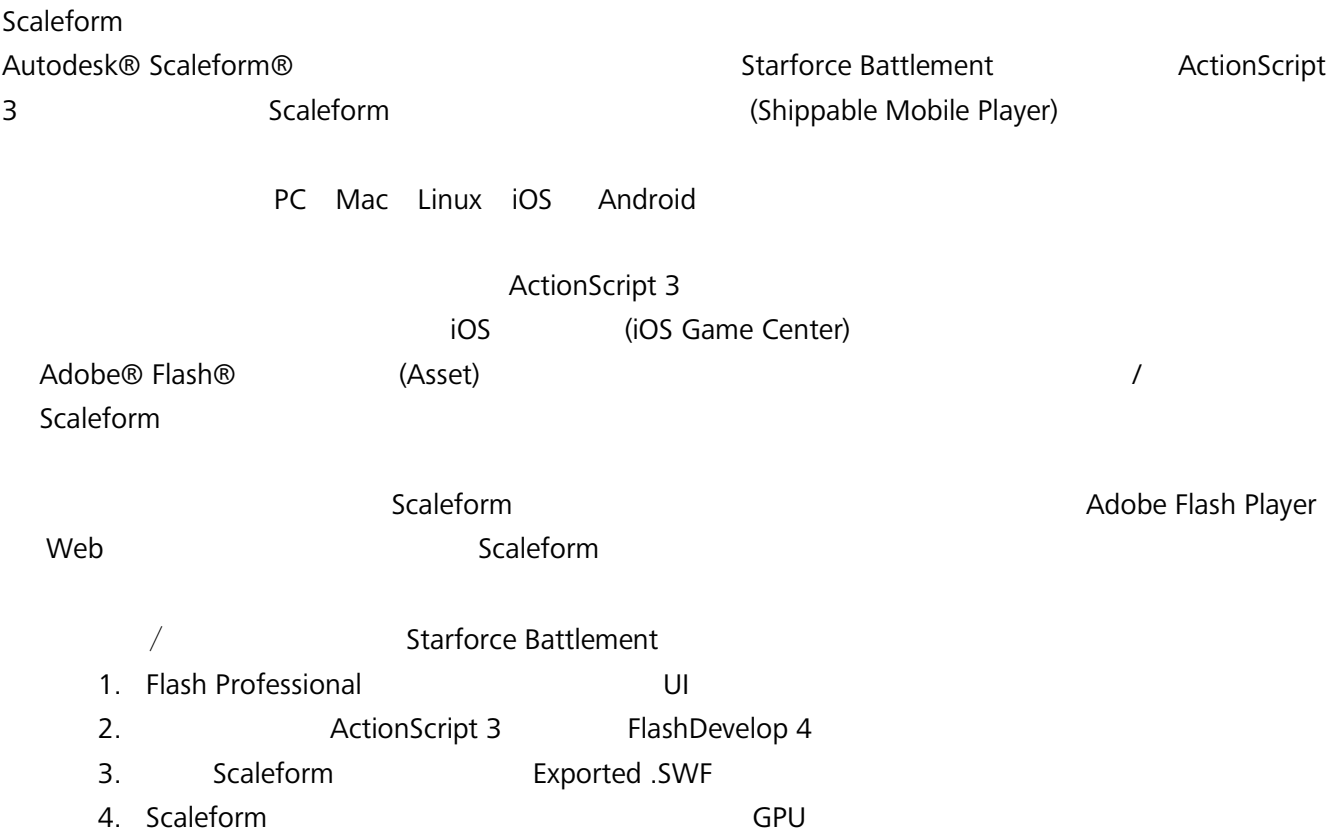
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How to Contact Autodesk Scaleform:

Document	Mobile Game Kit Overview
Address	Autodesk Scaleform Corporation 6305 Ivy Lane, Suite 310 Greenbelt, MD 20770, USA
Website	www.scaleform.com
Email	info@scaleform.com
Direct	(301) 446-3200
Fax	(301) 446-3199

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1



1.1 Starforce Battlement



图 1 : Starforce – 1

Starforce Battlement

(Tower Defense Game)

(Credit) / (Unit/Tower)

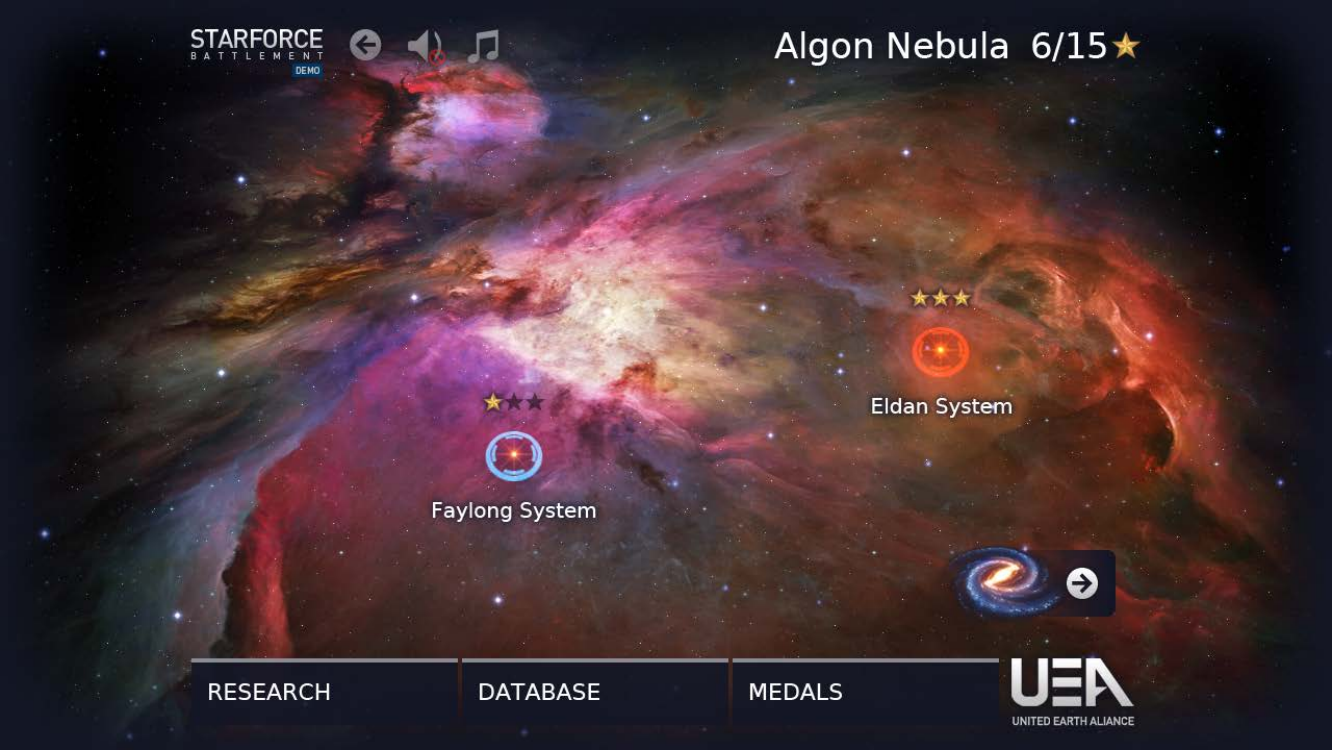


图 2 : Starforce –

Starforce Battlement

/

2012	5	iOS App Store	Starforce Battlement	iOS
		Scaleform	iOS	

1.2 特性

Starforce Battlement

Scaleform

- 1.
2. GPU
3. ActionScript 3
- 4.
- 5.
- 6.
7. iOS
8. (Orientation Lock) / (Orientation Handling)

 ActionScript 3

- 1.
- 2.
3. iOS
- 4.

2

2.1

- *Bin/Data/AS3/Kits/StarforceTD/* - ActionScript 3 Flash
- *Projects/Win32 /{Msvc80, Msvc90, or Msvc10}/Gfx 4.3 SDK / –* Visual Studio
Windows FxPlayer Visual Studio 2005/2008/2010
Starforce.
- *Projects/iPhone/Xcode4/Gfx 4.3 iPhone SDK –* Xcode 4 iOS
(Shippable Mobile Player) Starforce.
- *LocalApps\StarforceTD\Android –* make Scaleform
Eclipse “Shippable Mobile Player” Android
StarforceTD

Bin/Data/AS3/Kits/ StarforceTD Windows StarforceTD.exe
Windows Scaleform SDK
FxPlayer / FxMobilePlayer

Windows Debugging “Working directory”
Bin/Data/AS3/Kits/StarforceTD “Command line arguments”
StarforceTD.swf

iOS Xcode iTunes SWF

2.2

2.2.1

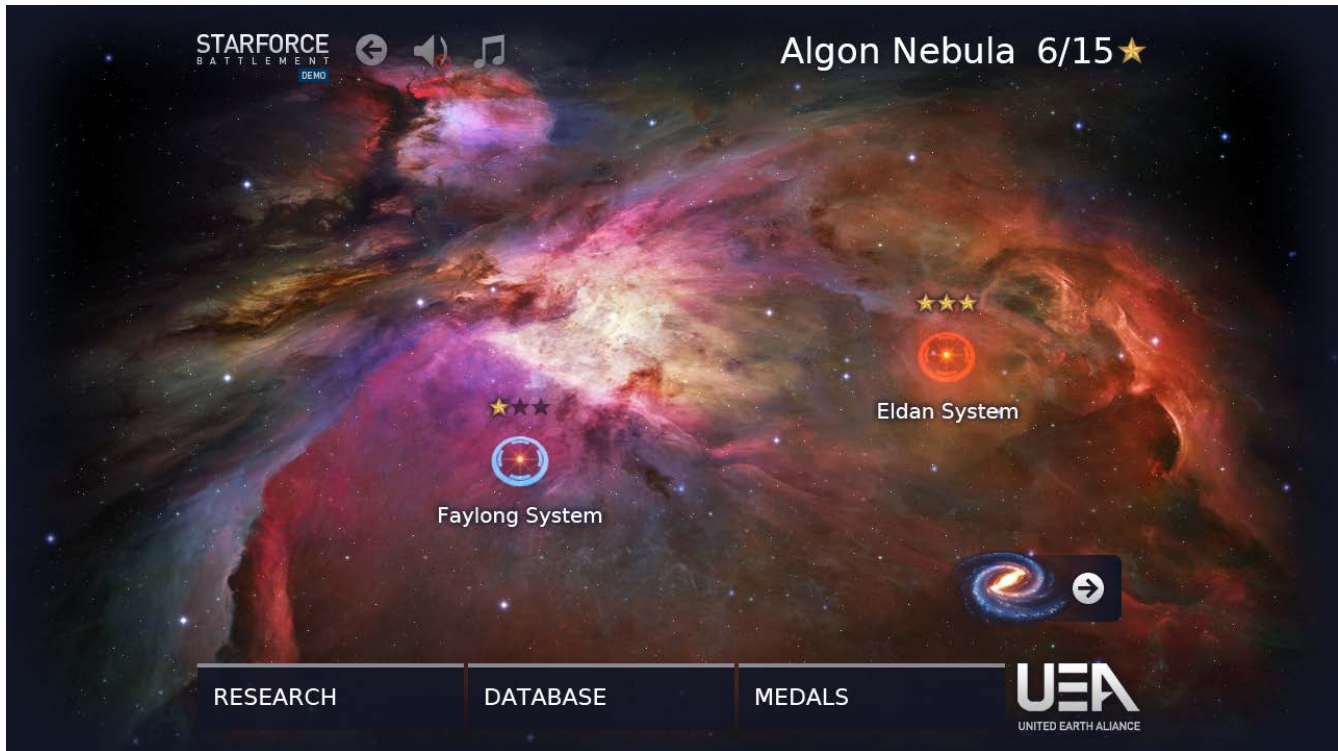


图 3 : Starforce – 主菜单

StarforceTD.as	StarforceTD.fla	Document Class
MainMainState	MainMainState	StarforceMenu.swf StarforceMenu.swf
	ActionScript 3	

/ (galaxy/nebula)

2D



图 4 : Starforce –

Data View) (MissionDataView.as) (Mission
View) "X" (Enter Mission)
(Mission Select

2.2.2 HUD



图 5 : Starforce -

Starforce Battlement

StarforceTD.swf
LevelGameState

StarforceTD.swf

/

-- "Rush Wave"

--

HUD

1

2

3

(Health)

HUD

HUD

HUD

(Orbital Strike)

(Deploy Mercenaries)

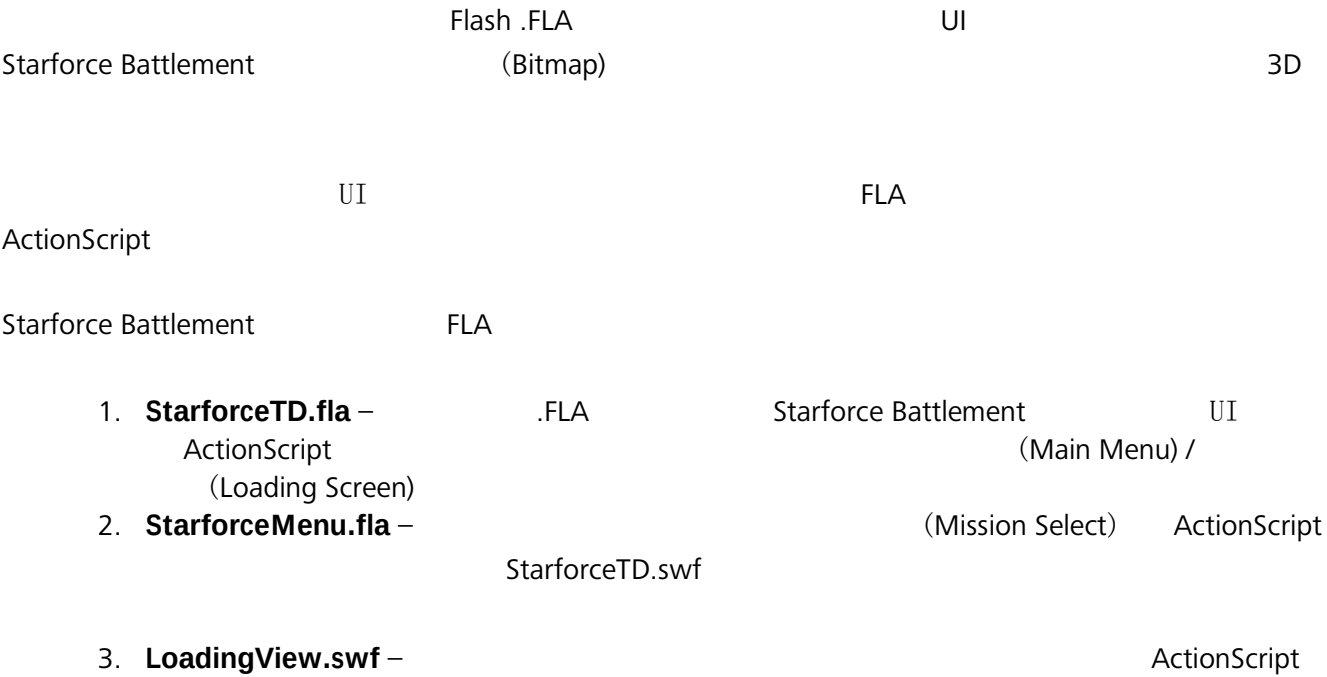
HUD

/

/

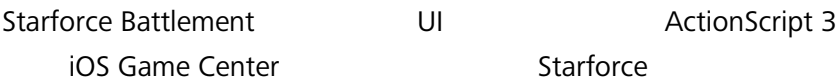
3

3.1 Flash

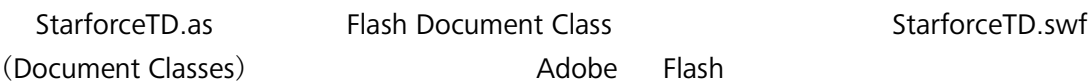


3.2 ActionScript

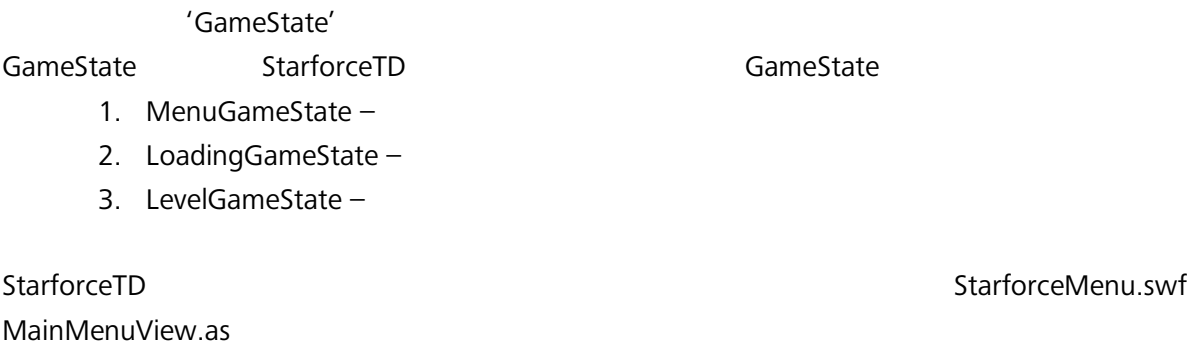
3.2.1



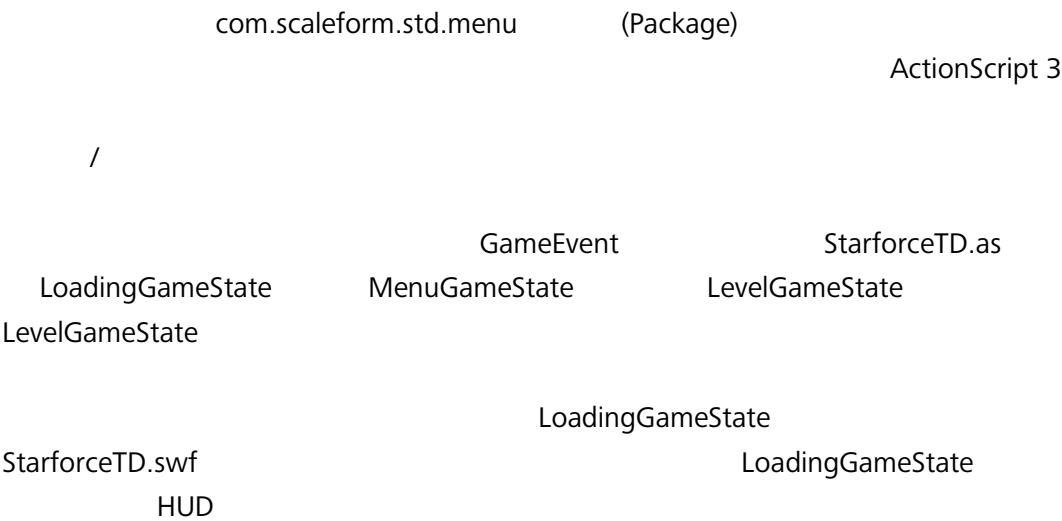
3.2.1.1



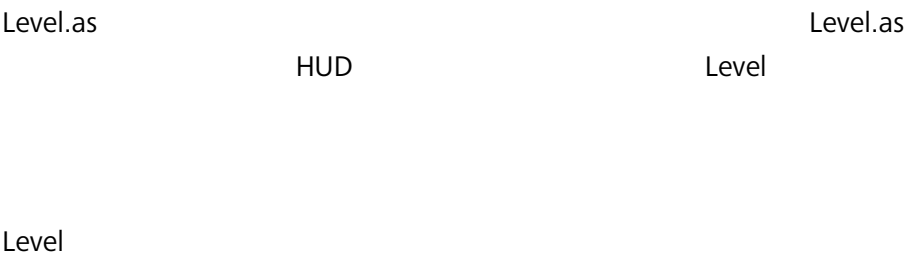
3.2.1.2 游戏状态



3.2.1.3



3.2.1.4



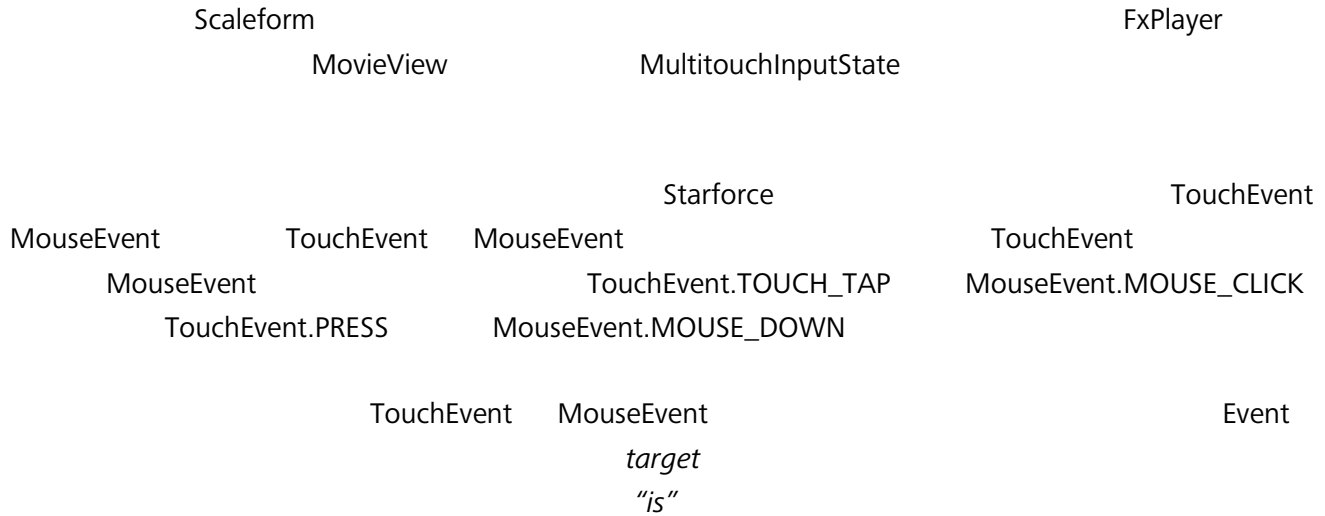
Flash Professional ActionScript (Symbol)
 ActionScript ActionScript
 Starforce Battlement
 ActionScript Adobe Flash
 TowerMech StarforceTD fla Flash Library TowerMech
 com.scaleform.std.entities.TowerMech ActionScript ActionScript
 Level.as
 HUD (Tower) (Unit)
 (Level) AttackEvent AttackEvents

3.2.1.5

com.scaleform.std.data
 GameData
 GameData
 ActionScript XML JSON
 Scaleform 4.0 XML

3.2.1.6

ActionScript
 Scaleform / ActionScript



```

protected function onSetRallyClick( e:Event ):void {
    var point:Point;
    if (e is MouseEvent) {
        var me:MouseEvent = e as MouseEvent;
        point = new Point( me.stageX, me.stageY );
    }
    else {
        var te:TouchEvent = e as TouchEvent;
        point = new Point( te.stageX, te.stageY );
    }
}

```

3.2.2 ActionScript

(Package)	/	ActionScript	ActionScript ‘ ,
		“com.scaleform.std”	
1. abilities	–		
2. controls	–	UI	
3. core	–	(State Flow)	/
4. data	–		
5. entities	–	游戏的数据定义，包括玩家、实体、关卡和能力。	
6. events	–	UI	
7. fx –			
8. hud–	HUD		
9. levels	–	Level	

10. loading	–	(Loading Screen)		
11. menu	–		UI	
12. system	–			ActionScript
13. utils	–			

3.2.3

3.2.3.1 /

com.scaleform.std.system.SaveSystem.as

ActionScript	SharedObject	SharedObject	ActionScript 3
(Primitive Type)	(Array)	Scaleform	SharedObject
SharedObject			
SharedObject	"sb_player_data"		

```
// Local reference to the galaxy progress. Populated from the PlayerData.
public static var galaxyProgress:Array = null;
// Constant String for the property of the SharedObject we'll modify.
protected static const GALAXY_PROGRESS_SO_PROP:String = "galaxyProgress";
// Reference to the SharedObject once we retrieve it.
protected static var _playerDataSO:SharedObject = null;

public static function load( pd:PlayerData ):void {
    _playerDataSO = SharedObject.getLocal("sb_player_data");
    galaxyProgress = _playerDataSO.data[ GALAXY_PROGRESS_SO_PROP ];
    pd.setGalaxyProgress( galaxyProgress );
}

public static function save( pd:PlayerData ):void {
    // Write PlayerData into SharedObject.
    _playerDataSO = SharedObject.getLocal("sb_player_data");
    galaxyProgress = pd.getGalaxyProgress();
    _playerDataSO.data[ GALAXY_PROGRESS_SO_PROP ] = galaxyProgress;
    _playerDataSO.flush();
}
```

SharedObject
SharedObject

SharedObject

SharedObject

http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/net/SharedObject.html

3.2.3.2

com.scaleform.std.system.SoundSystem

SoundEvent

.mp3

SoundEvent stage

```
dispatchEvent( new SoundEvent( SoundEvent.PLAY_SOUND, true, true, "infantry_attack.mp3",  
                                1, false ) );
```

SoundEvent

SoundEvent

```
dispatchEvent( new SoundEvent( SoundEvent.MUTE_SOUNDFX, true, true );
```